



哈尔滨工业大学
HARBIN INSTITUTE OF TECHNOLOGY

生成模型

- 自编码器 (AutoEncoder)
 - Denoising AutoEncoder (DAE)、Variational AutoEncoder (VAE)、Vector Quantization VAE (VQ-VAE)
- 归一化流/标准化流 (Normalizing Flow)
- 扩散模型 (Diffusion Model)
- 自回归模型 (Visual Autoregressive Model)

规格严格 功夫到家

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AutoEncoder

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自编码器

AutoEncoder

As close as possible (reconstruction)

3 x 3 Find pattern, and then compress 2 dim 3 x 3

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去噪自编码器

Denoising AutoEncoder

As close as possible

Add noises

3 x 3 NN Encoder Vector NN Decoder 3 x 3

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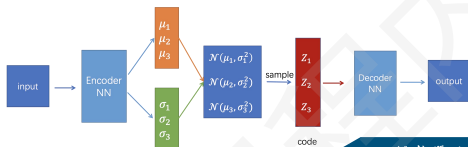
自编码器 VS 去噪自编码器

维度	AE	DAE
输入	x	$\bar{x} = x + n$
目标	x	x
学习目标	恒等映射	学习鲁棒的特征表达
隐空间特性	可能导致平凡解	强制编码器捕获更抽象、对扰动不敏感的特征
主要用途	降维、特征压缩	数据去噪、鲁棒特征学习

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变 用一个“近似的、可解的分布”去代替“真实的、不可解的分布”，这种“用简单分布近似复杂分布”的方法，就叫变分法。

- 本质上是判别模型，只关心将输入 x 映射到 $\hat{x}$ ，不关心整体分布 $P(x)$
- 变分自编码器
 - 本质上是生成模型，引入概率分布 $P(x)$ ，从而能够生成新的、相似样本



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变分自编码器

- 给定图像 x ，隐变量 z 的分布

$$p(z|x) = \frac{p(x|z)p(z)}{p(x)}$$

然而 $p(x)$ 是数据真实分布（所有可能图片的概率），无法计算，导致 $p(z|x)$ 无法求解

- 解决思路
 - 使用近似分布 $q(z|x)$ 代替，并令 $q(z|x) \sim \mathcal{N}(\mu, \sigma^2)$

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自编码器/去噪自编码器 VS 变分自编码器

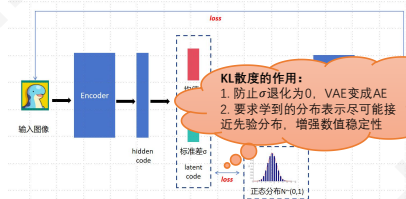
维度	AE/DAE	VAE
隐空间	确定的点	一个概率分布
编码器输出	一个特征向量 z	分布的参数 (μ, σ)
解码器输入	z	$z \sim \mathcal{N}(\mu, \sigma)$
损失函数	重建损失（例如 $\ x - \hat{x}\ _2^2$ ）	重建损失 + KL 散度
能否生成新数据	基本不能	能（相似样本）

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$z \sim \mathcal{N}(\mu, \sigma)$ 的梯度传导问题

- 变分自编码器的重参数化技巧

$$z \sim \mathcal{N}(\mu, \sigma) \rightarrow z = \mu + \sigma \cdot \epsilon, \epsilon \sim \mathcal{N}(0, 1)$$



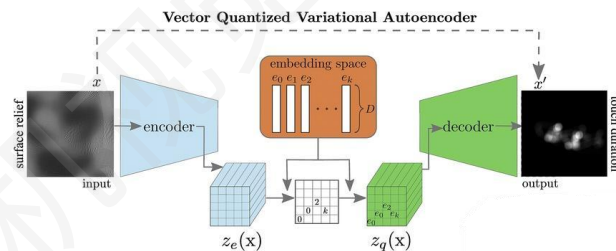
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VQ-VAE

- VAE 存在的问题
 - 平滑无结构的高斯分布难以拟合真实世界数据，导致重建时学习平均、模糊的结果
 - 重建损失与 KL 散度的妥协：具有重叠特征的类别（例如数字 1 和 7）直接平均输出
- 替代方案
 - 放弃连续假设，维护离散的、有限的隐变量
 - 对特定目标（例如人脸）有较好的效果

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VQ-VAE



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VQ-VAE 梯度计算与传导

➤ Straight-Through Estimator

- Code 匹配
`encoding_index = torch.argmax(distances, dim=1)`
`z_q = embedding(encoding_index)`
- 梯度传导
`z_q_st = z_e + (z_q - z_e).detach()`
- VQ 函数计算
`codebook_loss = F.mse_loss(z_q, z_e.detach())` # 通常权重较大快速学习
`commitment_loss = 0.25 F.mse_loss(z_q.detach(), z_e)` # 权重小稳定编码器
- 重建损失计算
`x_recon = decoder(z_q_st)`
`recon_loss = F.mse_loss(x_recon, x)`

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Normalizing Flow

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Normalizing Flow

➤ 核心思想：分布变换公式

- 假设有可逆变换 $x = f(z)$ ，且 $z = f^{-1}(x)$ ，那么有概率质量守恒
 $p_x(x) \cdot dx = p_z(z) \cdot dz$

- 两边除以 dx

$$p_x(x) = p_z(z) \cdot \left| \frac{dz}{dx} \right|$$

- 由于 $z = f^{-1}(x)$ ，所以 $\frac{dz}{dx} = (f^{-1})'(x)$

$$p_x(x) = p_z(f^{-1}(x)) \cdot |(f^{-1})'(x)|$$

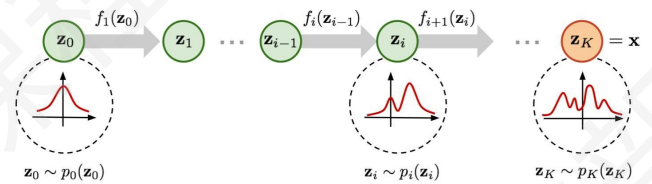
- 高维情况

$$p_y(y) = p_x(f^{-1}(y)) * |\det J_{f^{-1}}(y)|$$

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Normalizing Flow



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可逆网络

➤ Coupling Layer (耦合层) 设计 - 加性耦合层 (NICE)

- 将 D 维特征 x ，将其划分为 x_1 和 x_2 ，并做变换

$$\begin{aligned} y_1 &= x_1 \\ y_2 &= x_2 + m(x_1) \end{aligned}$$

- 其中， m 是任意网络，则逆运算可得

$$\begin{aligned} x_1 &= y_1 \\ x_2 &= y_2 - m(x_1) \\ &= y_2 - m(y_1) \end{aligned}$$

不改变尺寸以保证可逆

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可逆网络

➤ Coupling Layer (耦合层) 设计 - 仿射变换耦合层 (RealNVP)

- 将 D 维特征 x ，将其划分为 x_1 和 x_2 ，并做变换

$$\begin{aligned} y_1 &= x_1 \\ y_2 &= s(x_1) \circ x_2 + t(x_1) \end{aligned}$$

- 其中， s 和 t 是任意网络，则逆运算可得

$$\begin{aligned} x_1 &= y_1 \\ x_2 &= (y_2 - t(x_1)) / s(x_1) \\ &= (y_2 - t(y_1)) / s(y_1) \end{aligned}$$



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扩散模型

生成式建模

➤ 基于似然的生成模型



$$\text{VAE: } \log p_{\theta}(x^{(i)}) = D_{KL}(q_{\phi}(z|x^{(i)})||p_{\theta}(z|x^{(i)})) + \mathcal{L}(\theta, \phi; x^{(i)})$$

$$\text{Normalizing Flow: } \log p_{\theta}(x) = \log p_{\theta}(z) + \log |\det(dz/dx)|$$

$$\text{GAN: } \min_G \max_D V(D, G) = \mathbb{E}_{z \sim p_{\text{data}}(z)} [\log D(z)] + \mathbb{E}_{z \sim p_G(z)} [\log(1 - D(G(z)))]$$

Kingma, D.P., and Max Welling. Auto-Encoding Variational Bayes. ICLR 2014. Eqs. (1)
Kingma, D.P., and P. Dhariwal. Glow: Generative Flow with Invertible 1 + 1 Convolutions. NeurIPS 2018. Eqs. (6)
Goodfellow, Ian, et al. Generative Adversarial Nets. NeurIPS 2014. Eqs. (1)

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极大似然 (MLE)

➤ 基于似然的生成模型：最大化 $\log p_{\theta}(x)$ 

$$\theta^* = \arg \max_{\theta} [\log p_{\theta}(x)]$$

$$\theta_{t+1} = \theta_t + \alpha \cdot \bar{\alpha}_{x \sim p_{\text{data}}(x)} [\nabla_{\theta} \log p_{\theta}(x)]$$

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基于得分匹配的模型 (Score-based Models)

➤ 与传统极大似然方法相比，直接优化 p_{data} 

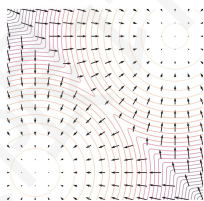
Traditional MLE	Score-based Models (SBMs)
$\theta^* = \arg \max_{\theta} [\log p_{\theta}(x)]$	$s_{\theta}(x) \approx s(x) \triangleq \nabla_x \log p_{\text{data}}(x) \in \mathbb{R}^d$
$\theta_{t+1} = \theta_t + \alpha \cdot \bar{\alpha}_{x \sim p_{\text{data}}(x)} [\nabla_{\theta} \log p_{\theta}(x)]$	$x' = x + \alpha \cdot s(x)$

Hyvärinen, Aapo. Estimation of non-normalized statistical models using score matching. JMLR 2005.

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基于得分匹配的模型 (Score-based Models)

➤ 显式 SBMs - 直接计算与真实分布的差异



A Mixture of Two Gaussians

Vector field: Score function $s_{\theta}(x)$; Contours: Density Function

Score-based Models (SBMs)
$s_{\theta}(x) \approx s(x) \triangleq \nabla_x \log p_{\text{data}}(x) \in \mathbb{R}^d$
$x' = x + \alpha \cdot s(x)$

$$J(\theta) = \bar{\alpha}_{p_{\text{data}}(x)} [\|s_{\theta}(x) - s(x)\|_2^2]$$

$$= \bar{\alpha}_{p_{\text{data}}(x)} [\|s_{\theta}(x) - \nabla_x \log p_{\text{data}}(x)\|_2^2]$$

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基于得分匹配的模型 (Score-based Models)

➤ 显式 SBMs - 直接计算与真实分布的差异

- 核心思想
 - 一步一步接近 $p_{\text{data}}(x)$
 - 引入部分随机性

- 两个问题
 - $s(x) \triangleq \nabla_x \log p_{\text{data}}(x)$ 无法获取
 - 无法保证数据未覆盖区间的泛化

SBMs with Langevin Dynamics
$x_{t+1} = x_t + \alpha_t \cdot s_{\theta_t}(x_t) + \sqrt{2\alpha_t} \cdot z(x)$
$z \sim N(0, I)$

$$J(\theta) = \bar{\alpha}_{p_{\text{data}}(x)} [\|s_{\theta}(x) - s(x)\|_2^2]$$

$$= \bar{\alpha}_{p_{\text{data}}(x)} [\|s_{\theta}(x) - \nabla_x \log p_{\text{data}}(x)\|_2^2]$$

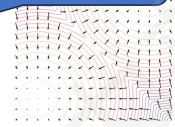
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隐式得分匹配 (ISM)

$$\begin{aligned}
 J(\theta) &= \bar{\alpha}_{p_{\text{data}}(\mathbf{x})} \left[\left\| \mathbf{s}_{\theta}(\mathbf{x}) - \mathbf{s}(\mathbf{x}) \right\|_2^2 \right] \\
 &= \bar{\alpha}_{p_{\text{data}}(\mathbf{x})} \left[\left\| \mathbf{s}_{\theta}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \right] \\
 &\Downarrow \\
 J_1(\theta) &= \bar{\alpha}_{p_{\text{data}}(\mathbf{x})} \left[\frac{1}{2} \left\| \mathbf{s}_{\theta}(\mathbf{x}) - \frac{\partial \log p_{\text{data}}(\mathbf{x})}{\partial \mathbf{x}} \right\|_2^2 \right] \\
 &= \bar{\alpha}_{p_{\text{data}}(\mathbf{x})} \left[\frac{1}{2} \left\| \mathbf{s}_{\theta}(\mathbf{x}) \right\|_2^2 + \sum_{i=1}^d \frac{\partial s_{\theta,i}(\mathbf{x})}{\partial x_i} \right] + C_1 \\
 &= \bar{\alpha}_{p_{\text{data}}(\mathbf{x})} \left[\frac{1}{2} \left\| \mathbf{s}_{\theta}(\mathbf{x}) \right\|_2^2 + \text{tr}(\nabla_{\mathbf{x}} \mathbf{s}_{\theta}(\mathbf{x})) \right]
 \end{aligned}$$

本页跳过:
核心思想是通过隐式得分匹配避免计算海森矩阵



$$\lim_{\|\mathbf{x}\| \rightarrow \infty} p_{\text{data}}(\mathbf{x}) s_{\theta}(\mathbf{x}) = 0$$

A constant irrelevant to θ

Hyvriinen, Aapo. Estimation of non-normalized statistical models using score matching. JMLR 2005.

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隐式得分匹配 (ISM)

$$\begin{aligned}
 J_1(\theta) &= \bar{\alpha}_{p_{\text{data}}(\mathbf{x})} \left[\frac{1}{2} \left\| \mathbf{s}_{\theta}(\mathbf{x}) - \frac{\partial \log p_{\text{data}}(\mathbf{x})}{\partial \mathbf{x}} \right\|_2^2 \right] \\
 &= \bar{\alpha}_{p_{\text{data}}(\mathbf{x})} \left[\frac{1}{2} \left\| \mathbf{s}_{\theta}(\mathbf{x}) \right\|_2^2 + \sum_{i=1}^d \frac{\partial s_{\theta,i}(\mathbf{x})}{\partial x_i} \right] + C_1 \\
 &= \bar{\alpha}_{p_{\text{data}}(\mathbf{x})} \left[\frac{1}{2} \left\| \mathbf{s}_{\theta}(\mathbf{x}) \right\|_2^2 + \text{tr}(\nabla_{\mathbf{x}} \mathbf{s}_{\theta}(\mathbf{x})) \right] \\
 &= \bar{\alpha}_{p_{\text{data}}(\mathbf{x})} \left[\frac{1}{2} \left\| \mathbf{s}_{\theta}(\mathbf{x}) \right\|_2^2 + \mathbf{v}^T \nabla_{\mathbf{x}} \mathbf{s}_{\theta}(\mathbf{x}) \mathbf{v} \right]
 \end{aligned}$$

$$\lim_{\|\mathbf{x}\| \rightarrow \infty} p_{\text{data}}(\mathbf{x}) s_{\theta}(\mathbf{x}) = 0$$

A constant irrelevant to θ

Second-order gradient

Vector-jacobian product (VJP),
computationally efficient in PyTorch

$$\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \in \mathbb{R}^d$$

Song, Yang, et al. Sliced score matching: A scalable approach to density and score estimation. UAI 2020.

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Recall DAE

- Change the input to the one with noise / partially masked
- BERT, MAE, ...

$$\begin{aligned}
 \hat{\mathbf{x}} &= \mathbf{x} + \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \\
 \arg \min_{\theta} \bar{\alpha}_{p(\mathbf{x}, \mathbf{z})} \left[\left\| f(\hat{\mathbf{x}}) - \mathbf{x} \right\|_2^2 \right]
 \end{aligned}$$

Vincent, Pascal, et al. Extracting and composing robust features with denoising autoencoders. ICML 2008.

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去噪得分匹配 (DSM)

- Adding AWGN to the data

$$\begin{aligned}
 p_{\text{data}}^{\sigma}(\tilde{\mathbf{x}}) &= \int p_{\text{data}}^{\sigma}(\tilde{\mathbf{x}}, \mathbf{x}) d\mathbf{x} \\
 p_{\text{data}}^{\sigma}(\tilde{\mathbf{x}}, \mathbf{x}) &= p_{\mathbf{x}}^{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) \cdot p_{\text{data}}(\mathbf{x})
 \end{aligned}$$

Gaussian Distribution spans over entire $\mathbb{R}^d$, covering much more regions than $p_{\text{data}}(\mathbf{x})$.

- Objective

$$\begin{aligned}
 J_D(\theta) &= \bar{\alpha}_{p_{\text{data}}^{\sigma}(\tilde{\mathbf{x}})} \left[\left\| \mathbf{s}_{\theta}(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log p_{\text{data}}^{\sigma}(\tilde{\mathbf{x}}) \right\|_2^2 \right] \\
 &= \bar{\alpha}_{p_{\text{data}}^{\sigma}(\tilde{\mathbf{x}}, \mathbf{x})} \left[\left\| \mathbf{s}_{\theta}(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log p_{\mathbf{x}}^{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) \right\|_2^2 \right]
 \end{aligned}$$

$$\text{For Gaussian, } \nabla_{\tilde{\mathbf{x}}} \log p_{\mathbf{x}}^{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) = -\frac{1}{\sigma^2} (\tilde{\mathbf{x}} - \mathbf{x})$$

Parzen Window

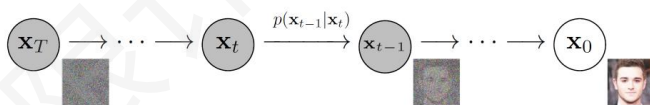
Vincent, Pascal. A connection between score matching and denoising autoencoders. Neural computation 2011.

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扩散模型

- 简单分布 (e.g., 高斯) $\rightarrow$ 复杂分布 (e.g., 真实数据)



$$p(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad p(\mathbf{x}_T) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad p(\mathbf{x}_{t-1} | \mathbf{x}_t) \text{ unknown}$$

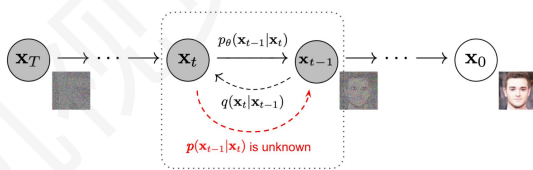
Ho, Jonathan, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. NeurIPS 2020.

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扩散模型

- 框架



$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad \text{where } \beta_t \in (0, 1)_{t=1}^T$$

Ho, Jonathan, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. NeurIPS 2020.

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扩散模型

分布性质

- Expressiveness of the reverse process (See the two references below for proof)

hyperparameters, and expressiveness of the reverse process is ensured in part by the choice of Gaussian conditionals in $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$, because both processes have the same functional form when β_t are small [53]. A notable property of the forward process is that it admits sampling $\mathbf{x}_t$ at an

In other words, if β_t is small enough, $p(\mathbf{x}_{t-1}|\mathbf{x}_t)$ will also be Gaussian.

Feller, William. On the theory of stochastic processes, with particular reference to applications. Berkeley Symposium on Mathematical Statistics and Probability. University of California Press, 1949.
Sohl-Dickstein, Jascha, et al. Deep unsupervised learning using nonequilibrium thermodynamics. ICML 2015.

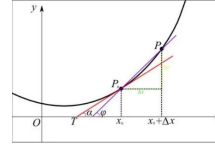
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扩散模型

分布性质

- Expressiveness of the reverse process: *An intuitive illustration*



$$f'_-(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0) - f(x_0 - \Delta x)}{\Delta x}$$

$$f'_+(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

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扩散模型

推导

- Closed-form for the t -th step, $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$

$$\begin{aligned} \mathbf{x}_t &= \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \mathbf{z}_{t-1} \\ &= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}} \mathbf{z}_{t-2}) + \sqrt{1 - \alpha_t} \mathbf{z}_{t-1} \\ &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t-1})} \mathbf{z}_{t-2} + \sqrt{1 - \alpha_t} \mathbf{z}_{t-1} \\ &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\mathbf{z}}_{t-2} \\ &= \dots \\ &= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \mathbf{z} \end{aligned}$$

$$\mathcal{N}(0, \sigma_1^2 \mathbf{I}) + \mathcal{N}(0, \sigma_2^2 \mathbf{I}) \Rightarrow \mathcal{N}(0, (\sigma_1^2 + \sigma_2^2) \mathbf{I})$$

Thus, $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$

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扩散模型

推导

- $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$ $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$
- $p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)$, where $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) \triangleq \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$

问题变为求解 $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)$

目标: $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) \rightarrow p(\mathbf{x}_{t-1} | \mathbf{x}_t)$

问题: $p(\mathbf{x}_{t-1} | \mathbf{x}_t)$ 没有显式表示

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扩散模型

在训练过程引入 $\mathbf{x}_0$

$$\begin{aligned} p(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) &\triangleq \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0, \tilde{\beta}_t), \tilde{\beta}_t \mathbf{I}) \\ &= q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0) \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)} \\ &\propto \exp\left(-\frac{1}{2} \left(\frac{(\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_{t-1})^2}{\beta_t} + \frac{(\mathbf{x}_{t-1} - \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0)^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)^2}{1 - \bar{\alpha}_t} \right)\right) \\ &= \exp\left(-\frac{1}{2} \left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) \mathbf{x}_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{2\sqrt{\bar{\alpha}_t}}{1 - \bar{\alpha}_t} \mathbf{x}_0 \right) \mathbf{x}_{t-1} + C(\mathbf{x}_t, \mathbf{x}_0) \right)\right) \end{aligned}$$

规格严格 功夫到家



扩散模型

在训练过程引入 $\mathbf{x}_0$

$$\begin{aligned} p(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) &\triangleq \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0, \tilde{\beta}_t), \tilde{\beta}_t \mathbf{I}) \\ &\propto \exp\left(-\frac{1}{2} \left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) \mathbf{x}_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{2\sqrt{\bar{\alpha}_t}}{1 - \bar{\alpha}_t} \mathbf{x}_0 \right) \mathbf{x}_{t-1} + C(\mathbf{x}_t, \mathbf{x}_0) \right)\right) \end{aligned}$$

由高斯密度函数推导,

$$\begin{aligned} \tilde{\beta}_t &= 1 / \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) = 1 / \left(\frac{\alpha_t(1 - \bar{\alpha}_{t-1})}{\beta_t(1 - \bar{\alpha}_{t-1})} + \frac{\beta_t}{\beta_t(1 - \bar{\alpha}_{t-1})} \right) = \frac{\beta_t(1 - \bar{\alpha}_{t-1})}{\alpha_t - \bar{\alpha}_{t-1} + \beta_t} = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t \\ \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) &= \left(\frac{\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_t}}{1 - \bar{\alpha}_t} \mathbf{x}_0 \right) / \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) = \frac{\sqrt{\alpha_t(1 - \bar{\alpha}_{t-1})}}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_t} \beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 \end{aligned}$$

规格严格 功夫到家



扩散模型

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \mathbf{z} \Rightarrow \mathbf{x}_0 = \frac{1}{\sqrt{\alpha_t}} (\mathbf{x}_t - \sqrt{1 - \alpha_t} \mathbf{z}_t)$$

$$\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0$$

$$= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \frac{1}{\sqrt{\alpha_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \mathbf{z}_t)$$

$\mathbf{x}_0$ 能够被消除!

$$= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\beta_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \mathbf{x}_t - \frac{\beta_t\sqrt{1 - \bar{\alpha}_t}}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \mathbf{z}_t$$

$$= \frac{\alpha_t + \beta_t - \bar{\alpha}_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}} \mathbf{z}_t$$

$$= \frac{1}{\sqrt{\alpha_t}} (\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{z}_t)$$

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扩散模型

我们有

$$p(\mathbf{x}_{t-1} | \mathbf{x}_t) \triangleq \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t \mathbf{I})$$

$$\begin{cases} \tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}} (\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{z}_t) \\ \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t \end{cases}$$

作弊路径...

我们有 $p(\mathbf{x}_{t-1} | \mathbf{x}_t)$, 可以直接约束 $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) \rightarrow p(\mathbf{x}_{t-1} | \mathbf{x}_t)$

$\tilde{\beta}_t$ 已知, $\mathbf{x}_t$ 已知, 只需要学习预测 $\mathbf{z}_t$,

$$\mathcal{L} = \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_t} [\|\mathbf{z}_t - \mathbf{z}_\theta(\sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \mathbf{z}_t, t)\|^2], \text{ where } \mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

规格严格 功夫到家



扩散模型

原文推导

MLE by VLB

$$\begin{aligned} -\log p_\theta(\mathbf{x}_0) &\leq -\log p_\theta(\mathbf{x}_0) + D_{\text{KL}}(q(\mathbf{x}_{1:T} \| p_\theta(\mathbf{x}_{1:T} | \mathbf{x}_0))) \\ &= -\log p_\theta(\mathbf{x}_0) + \mathbb{E}_{\mathbf{x}_{1:T} \sim q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \frac{q(\mathbf{x}_{1:T} | \mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T}) / p_\theta(\mathbf{x}_0)} \right] \\ &= -\log p_\theta(\mathbf{x}_0) + \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T} | \mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} + \log p_\theta(\mathbf{x}_0) \right] \\ &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T} | \mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \end{aligned}$$

规格严格 功夫到家



扩散模型

原文推导

MLE by VLB

Split into three parts

$$\begin{aligned} \mathcal{L}_{\text{VLB}} &= \mathcal{L}_1 + \mathcal{L}_2 + \dots + \mathcal{L}_T \\ \mathcal{L}_1 &= D_{\text{KL}}(q(\mathbf{x}_1 | \mathbf{x}_0) \| p(\mathbf{x}_1 | \mathbf{x}_0)) \\ \mathcal{L}_t &= D_{\text{KL}}(q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0) \| p(\mathbf{x}_t | \mathbf{x}_t)) \text{ for } 1 \leq t \leq T \\ \mathcal{L}_t &= D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p(\mathbf{x}_{t-1} | \mathbf{x}_t)) \text{ for } 2 \leq t \leq T \\ \mathcal{L}_T &= -\log p_\theta(\mathbf{x}_T | \mathbf{x}_T) \end{aligned}$$

$$\begin{aligned} \mathcal{L}_{\text{VLB}} &= \mathbb{E}_{\mathbf{x}_0} \left[\log \frac{q(\mathbf{x}_{1:T} | \mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \\ &= \mathbb{E} \left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_0) \prod_{t=1}^T p_\theta(\mathbf{x}_t | \mathbf{x}_t)} \right] \\ &= \mathbb{E} \left[-\log p_\theta(\mathbf{x}_0) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_t | \mathbf{x}_t)} \right] \\ &= \mathbb{E} \left[-\log p_\theta(\mathbf{x}_0) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_t | \mathbf{x}_t)} \log \frac{q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_t | \mathbf{x}_t)} \right] \\ &= \mathbb{E} \left[-\log p_\theta(\mathbf{x}_0) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_t | \mathbf{x}_t)} \log \frac{q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_t | \mathbf{x}_t)} \right] \\ &= \mathbb{E} \left[-\log p_\theta(\mathbf{x}_0) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_t | \mathbf{x}_t)} \log \frac{q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_t | \mathbf{x}_t)} \right] \\ &= \mathbb{E} \left[\log \frac{q(\mathbf{x}_1 | \mathbf{x}_1)}{p_\theta(\mathbf{x}_1)} \sum_{t=2}^T \log \frac{q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_t | \mathbf{x}_t)} \log \frac{q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_t | \mathbf{x}_t)} \right] \\ &= \mathbb{E} \left[D_{\text{KL}}(q(\mathbf{x}_1 | \mathbf{x}_1) \| p_\theta(\mathbf{x}_1)) + \sum_{t=2}^T \mathbb{E}_{\mathbf{x}_0} [q(\mathbf{x}_t | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_t | \mathbf{x}_t)] - \log p_\theta(\mathbf{x}_0) \right] \end{aligned}$$

规格严格 功夫到家



扩散模型

原文推导

MLE by VLB

Split into three parts

Focus on $\mathcal{L}_t$

$$\begin{aligned} \tilde{\mu}_t &= \frac{1}{\sqrt{\alpha_t}} (\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{z}_t) \\ \mu_t(\mathbf{x}_t, t) &= \frac{1}{\sqrt{\alpha_t}} (\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{z}_t) \\ \mathbf{x}_{t+1} &\sim \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t \mathbf{I}) = \mathcal{N}(\frac{1}{\sqrt{\alpha_t}} (\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{z}_t), \tilde{\beta}_t \mathbf{I}) \\ \mathcal{L}_t &= \mathbb{E}_{\mathbf{x}_0} \left[\frac{1}{2\|\Sigma_\theta(\mathbf{x}_t, t)\|^2} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_t(\mathbf{x}_t, t)\|^2 \right] \\ &= \mathbb{E}_{\mathbf{x}_0} \left[\frac{1}{2\|\Sigma_\theta(\mathbf{x}_t, t)\|^2} \left\| \frac{1}{\sqrt{\alpha_t}} (\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{z}_t) - \frac{1}{\sqrt{\alpha_t}} (\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{z}_t(\mathbf{x}_t, t)) \right\|^2 \right] \\ &= \mathbb{E}_{\mathbf{x}_0} \left[\frac{\beta_t^2}{2\alpha_t(1 - \bar{\alpha}_t)\|\Sigma_\theta(\mathbf{x}_t, t)\|^2} \|\mathbf{z}_t - \mathbf{z}_\theta(\mathbf{x}_t, t)\|^2 \right] \\ &= \mathbb{E}_{\mathbf{x}_0} \left[\frac{\beta_t^2}{2\alpha_t(1 - \bar{\alpha}_t)\|\Sigma_\theta(\mathbf{x}_t, t)\|^2} \|\mathbf{z}_t - \mathbf{z}_\theta(\sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \mathbf{z}_t, t)\|^2 \right] \\ &= \mathbb{E}_{\mathbf{x}_0} [\|\mathbf{z}_t - \mathbf{z}_\theta(\sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \mathbf{z}_t, t)\|^2] \quad (\text{simplified}) \end{aligned}$$

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ADM: Diffusion Models Beat GANs

Combining classifier guidance $p_\theta, \phi(x_t | x_{t+1}, y) = Z p_\theta(x_t | x_{t+1}) p_\phi(y | x_t)$

ADM: ablated diffusion model

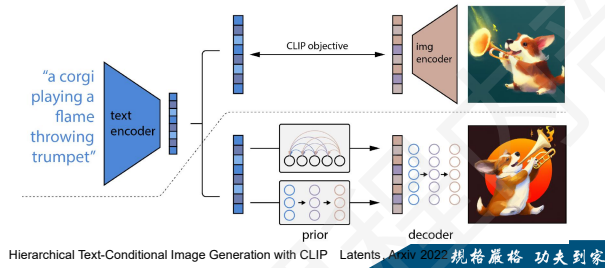
			Model	FID	↓	↑	Pre	Rec	Model	FID	↓	↑	Pre	Rec				
LSUN Bedroom 256 - 256																		
DCTransformer ¹ [42]															4.40	6.66	0.44	8.86
DiffGAN [23]															4.99	9.07	0.00	0.25
DiffGAN [23]															4.24	8.21	0.02	0.46
DiffGAN [23]															2.35	6.62	0.59	0.40
ADM (deep)															1.90	5.59	0.66	0.51
LSUN Horse 256 - 256																		
SquGAN2 [28]															1.84	6.46	0.63	0.49
ADM															1.05	5.84	0.69	0.55
ADM (deep)															2.57	6.81	0.71	0.55
LSUN Cat 256 - 256																		
DiffGAN [23]															17.1	12.4	0.51	0.40
SquGAN2 [28]															7.25	6.43	0.58	0.43
ADM (deep)															1.57	6.03	0.63	0.52
ADM															1.07	5.01	0.63	0.52
ImageNet 64 - 64																		
DiffGAN-deep [5]															1.00	3.96	8.79	0.40
DiffGAN [43]															2.02	4.70	0.74	0.63
ADM															1.00	3.77	0.71	0.63
ADM (deep)															2.07	4.30	0.74	0.63
ImageNet 128 - 128																		
DiffGAN-deep [5]															6.02	7.18	8.86	0.35
LSGAN ² [30]															5.26			
DiffGAN [43]															5.01	4.99	0.70	8.65
ADM-G (128 steps)															5.08	5.08	0.78	0.51
ADM-G															2.97	5.09	0.78	0.59
ImageNet 256 - 256																		
DCTransformer ¹ [42]															16.51	8.24	0.36	8.67
DiffGAN-deep [5]															14.11	17.36	0.36	0.57
DiffGAN [43]															12.26	5.42	0.70	0.62
DiffGAN [43]															11.90			
DiffGAN-deep [5]															6.95	7.36	8.87	0.28
ADM															10.84	4.02	0.69	0.63
ADM-G (128 steps)															5.84	5.32	0.81	0.49
ADM-G															4.09	4.28	0.82	0.52
ImageNet 432 - 432																		
DiffGAN-deep [5]															8.43	8.13	8.88	0.28
ADM															23.24	10.19	0.73	8.48
ADM-G (128 steps)															8.41	7.65	0.83	0.47
ADM-G															7.72	6.87	0.87	0.42

Diffusion Models Beat GANs on Image Synthesis. Arxiv 2021.

规格严格 功夫到家

DALL.E-2: unCLIP

- High quality 1024 × 1024 images



DALL.E-2: unCLIP



Imagen

- Abstract
 - Generic large language model, eg. T5, pretrained on text only corpora.
 - Text-guided super-res model

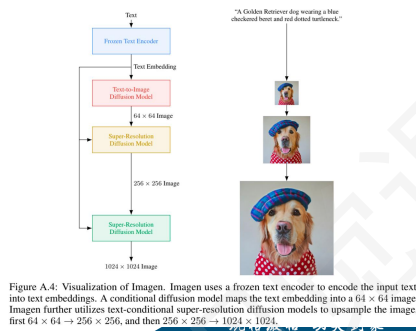
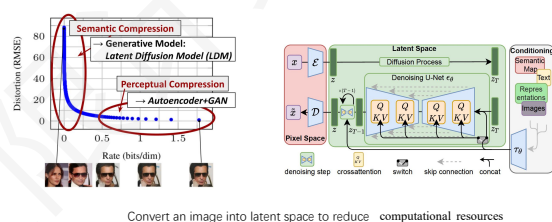


Imagen (Ours)

DALL.E 2 [54]

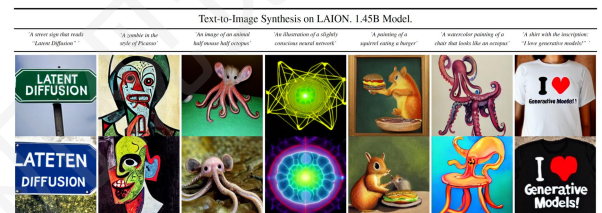


LDM (Latent Diffusion Models)



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LDM: Results



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Stable Diffusion

Stable Diffusion

Fork 4.6k Starred 31k

Stable Diffusion was made possible thanks to a collaboration with Stability AI and Runway and builds upon our previous work:

High-Resolution Image Synthesis with Latent Diffusion Models
Robin Rombach*, Andreas Blattmann*, Dominik Lorenz, Patrick Esser, Björn Ommer
CVPR '22 Oral | GitHub | arXiv | Project page



Stable Diffusion is a latent text-to-image diffusion model. Thanks to a generous compute donation from Stability AI and support from LAION, we were able to train a Latent Diffusion Model on 512x512 images from a subset of the LAION-5B database. Similar to Google's Imagen, this model uses a frozen CLIP ViT-L/14 text encoder to condition the model on text prompts. With its 860M UNet and 123M text encoder, the model is relatively lightweight and runs on a GPU with at least 10GB VRAM. See this section below and the model card.



集成现有Diffusion模型，直接调用

Visual Autoregressive Model

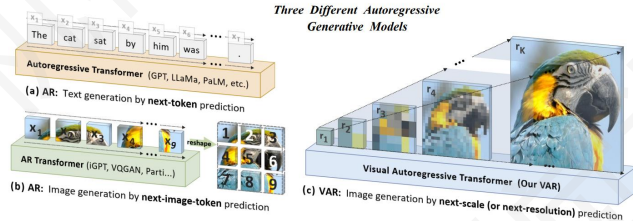
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Visual Autoregressive Modeling

Three Different Autoregressive Generative Models

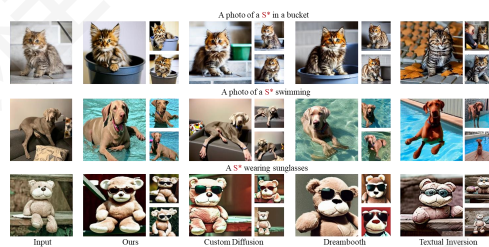


Visual Autoregressive Modeling: Scalable Image Generation via Next-Scale Prediction, NeurIPS 2024.

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定制化生成 Elite (ICCV '23)



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定制化生成 Elite (ICCV '23)

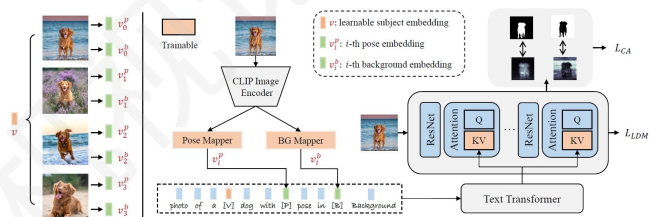
定制化图像生成方面，在生成性能至少相当的情况下，伪词生成速度是谷歌DreamBooth、英伟达Textual Inversion和Adobe的Custom Diffusion的1000倍以上。

Method	Text-alignment (↑)	Image-alignment (↑)	KID ($\times 10, \downarrow$)	Time (↓)
Textual Inversion	0.183	0.689	51.22	50 min
DreamBooth	0.249	0.827	4.48	30 min
Custom Diffusion	0.231	0.868	14.32	6 min
Ours	0.266	0.804	5.27	0.05 s

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定制化生成 Detex (AAAI '24)



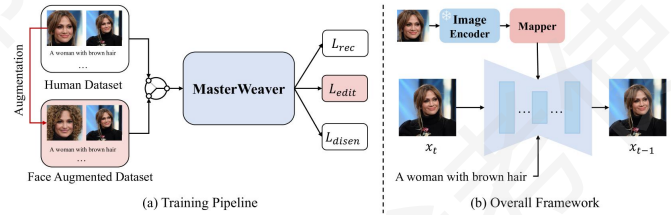
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定制化生成 Detex (AAAI '24)



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定制化生成 MasterWeaver (ECCV '24)



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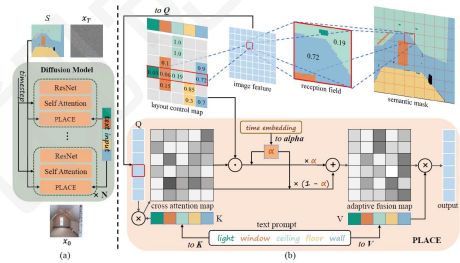


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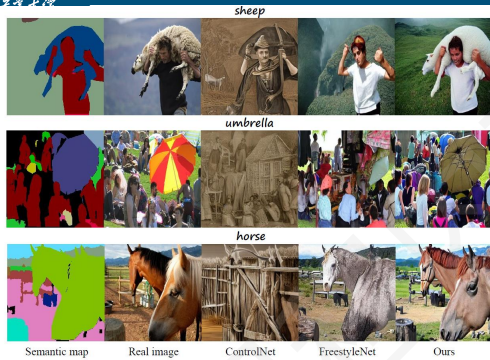


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空间布局可控生成 PLACE (CVPR '24)



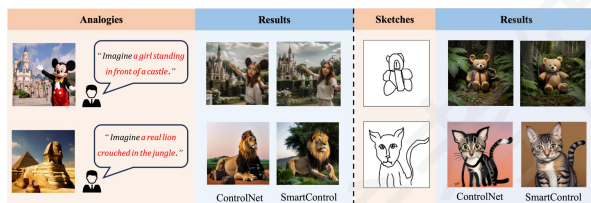
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夫到家

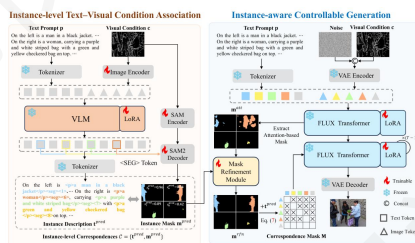
空间布局可控生成 SmartControl (ECCV'24)

相比像素级的精细控制，我们的方法只要求用户提供最关键的信息，可以更智能理解用户的意图。



规格严格 功夫到家

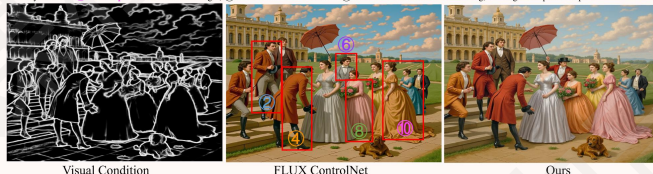
空间布局可控生成 InstanceControl (ArXiv'26)



规格严格 功夫到家

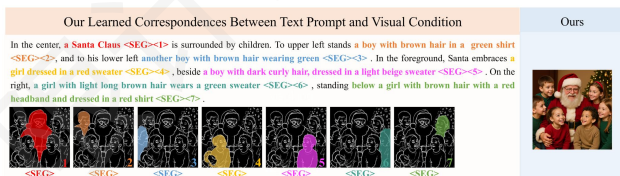
空间布局可控生成 InstanceControl (ArXiv'26)

... On the left stand three gentlemen, from bottom to top, ①one in an orange suit, ②the other is a red suit, and ③the third in a brown suit. To the lower right of the three gentlemen, ④a man in a red coat stands with his back to the viewer. In front of him is ⑤a woman in a silver dress. Behind her, ⑥a man in a red tuxedo and a black top hat holds ⑦a umbrella to shield her from the sun. Further to the right, ⑧a woman in an orange dress is crouching, holding a bouquet of flowers. To her right are two women: ⑨one in yellow and ⑩one in a pink dress. On the far right, ⑪a woman in a blue dress and ⑫a man in a black suit are conversing, creating a tranquil atmosphere. ...



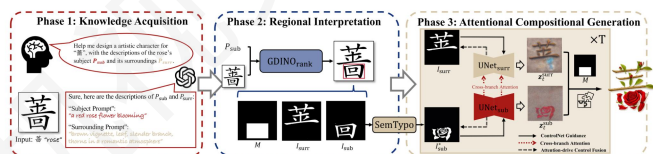
规格严格 功夫到家

空间布局可控生成 InstanceControl (ArXiv'26)



规格严格 功夫到家

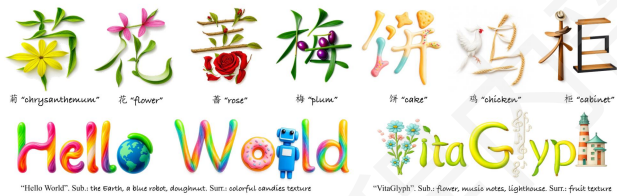
字体生成 VitaGlyph (WACV'26)



规格严格 功夫到家



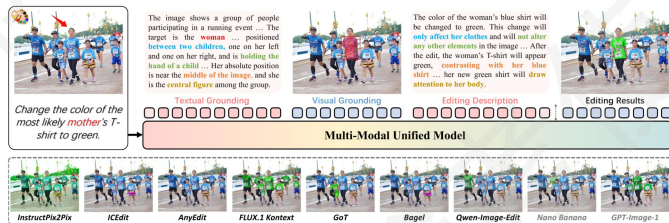
字体生成 VitaGlyph (WACV '26)



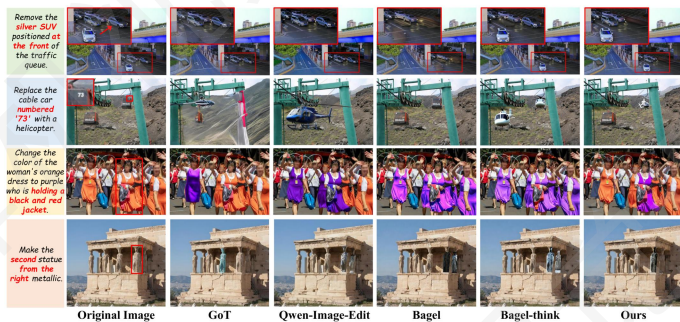
规格严格 功夫到家



空间精准编辑 InterCoG (ArXiv '26)



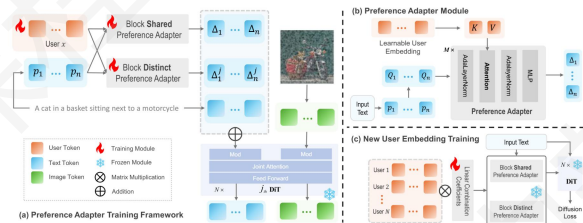
规格严格 功夫到家



规格严格 功夫到家



用户偏好对齐生成 Premier (CVPR '26)



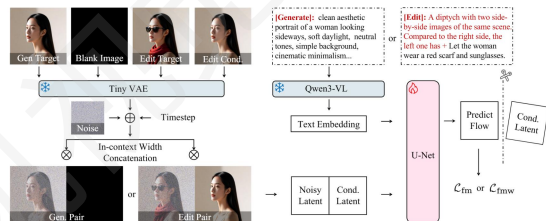
规格严格 功夫到家



User Preference Image

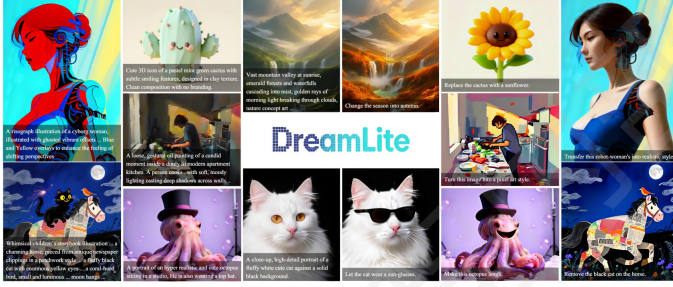


轻量化生成与编辑 DreamLite (ArXiv '26)



在小米14 (骁龙8 Gen3) 上实测<1秒完成1024×1024图像生成或编辑

规格严格 功夫到家



规格严格 功夫到家

Table 1: Evaluation on GenEval benchmark. "Param." denotes backbone parameters.

Model	Param.	Single Obj.	Two Obj.	Counting	Colors	Position	Color Attr.	Overall
FLUX.1-Dev [21]	12B	0.99	0.81	0.79	0.74	0.20	0.47	0.67
BAGEL [9]	7B	0.99	0.94	0.81	0.88	0.64	0.63	0.82
OmniGen2 [43]	4B	1.00	0.95	0.64	0.88	0.55	0.76	0.80
Z-Image [5]	6B	1.00	0.94	0.78	0.93	0.62	0.77	0.84
LongCat-Image [39]	6B	0.99	0.98	0.86	0.86	0.75	0.73	0.87
DeepGen1.0 [41]	2B	0.99	0.97	0.68	0.90	0.71	0.71	0.83
SANA-1.6B [46]	1.6B	0.99	0.80	0.62	0.89	0.27	0.43	0.67
Hunyuan-DiT [6]	1.5B	0.97	0.77	0.71	0.88	0.13	0.30	0.63
MEISSONIC [2]	1B	0.99	0.66	0.42	0.86	0.10	0.22	0.54
SnapGen [16]	0.38B	-	-	-	-	-	-	0.70
SnapGen++ [17]	0.4B	-	-	-	-	-	-	0.66
SANA-0.6B [46]	0.59B	0.98	0.78	0.59	0.86	0.22	0.40	0.64
Nitro-E-GRPO [38]	0.3B	1.00	0.88	0.63	0.86	0.32	0.48	0.70
Ours	0.39B	0.98	0.88	0.65	0.90	0.35	0.52	0.72

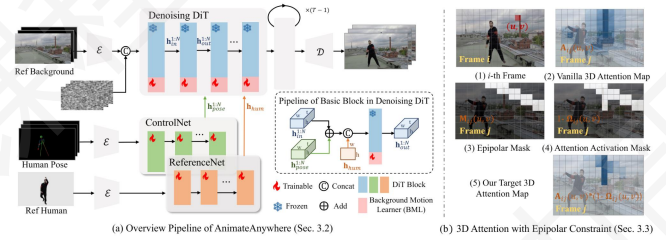
规格严格 功夫到家

Table 2: Evaluation on ImgEdit benchmark. "Param." denotes backbone parameters.

Model	Param.	Add	Remove	Replace	Adjust	BG	Style	Extract	Action	Hybrid	Overall
Kontext-Dev [23]	12B	4.04	2.92	4.28	3.89	4.00	4.43	2.71	4.52	3.08	3.76
BAGEL [9]	7B	3.81	3.16	3.85	3.59	3.39	4.51	1.58	4.25	2.67	3.42
OmniGen2 [43]	4B	3.57	3.20	3.74	3.06	3.57	4.81	1.77	4.68	2.52	3.44
Z-Image [5]	6B	4.40	4.13	4.57	4.14	4.14	4.85	4.30	4.50	3.63	4.30
LongCat-Image [39]	6B	4.66	4.68	4.75	4.64	4.35	4.81	3.90	4.83	3.80	4.49
DeepGen1.0 [41]	2B	4.52	4.43	4.42	4.09	4.38	4.48	1.75	4.57	3.61	4.03
EditMGT [8]	0.96B	2.78	2.89	3.34	3.29	3.05	4.17	1.56	2.94	2.01	2.89
VIBE [1]	1.6B	3.89	4.42	4.34	4.22	4.22	4.40	2.90	2.75	3.52	3.85
Ours	0.39B	4.59	4.27	4.32	4.46	4.28	4.42	2.70	4.73	3.24	4.11

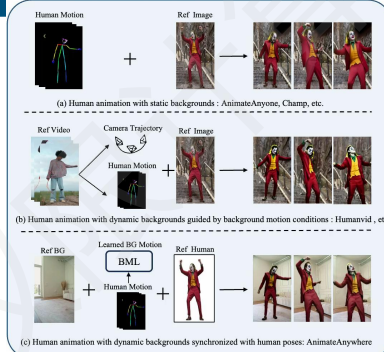
规格严格 功夫到家

动作/背景可控生成 AnimateAnywhere (ArXiv '25)

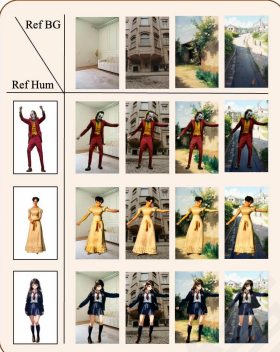


规格严格 功夫到家

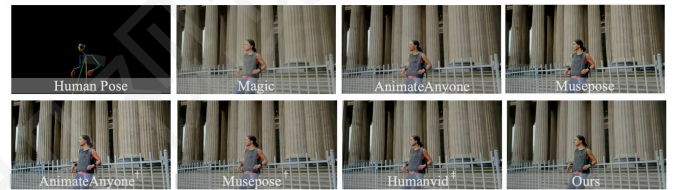
Comparison with Paradigms of Human Animation



The Results of AnimateAnywhere



动作/背景可控生成 AnimateAnywhere (ArXiv '25)

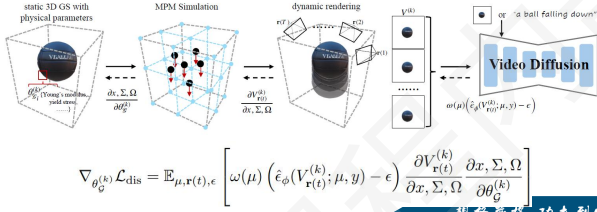


规格严格 功夫到家

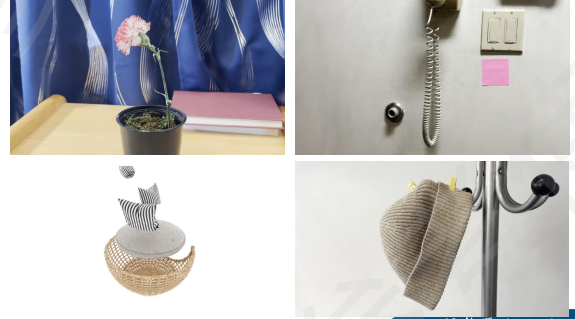


物理一致性视频生成 DreamPhysics (AAAI '25)

- 通过整合3D生成与表示、视频生成和物理运动方程，提出了物理机制驱动的视频生成（与MIT同期开源），以生成符合物理规律的视频



规格严格 功夫到家

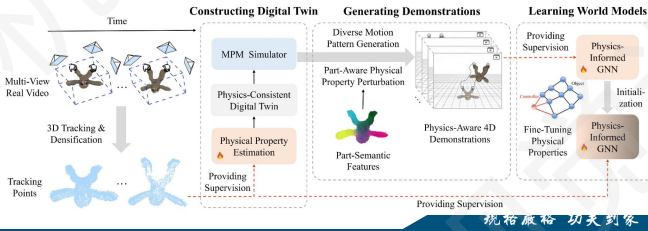


规格严格 功夫到家

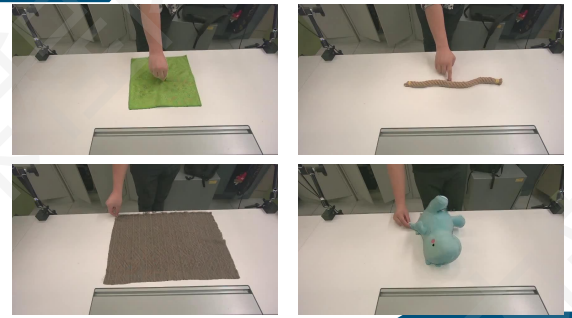


物理一致性视频生成 PhysWorld (ArXiv '25)

- 从视频中构建数字孪生，进而学习物理规律并构建世界模型，实现对未来帧的预测



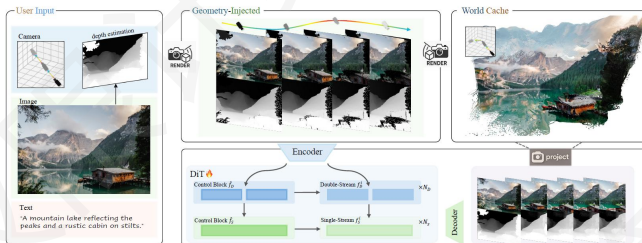
规格严格 功夫到家



规格严格 功夫到家



世界模型探索 Voyager (ToG '25)

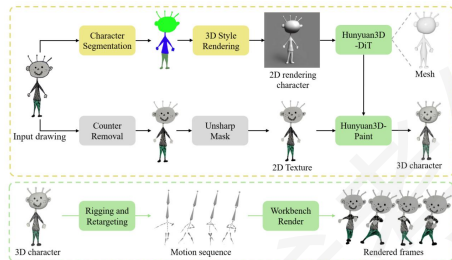


规格严格 功夫到家

Voyager: Long-Range and World-Consistent Video Diffusion for Explorable 3D Scene Generation

Harbin Institute of Technology

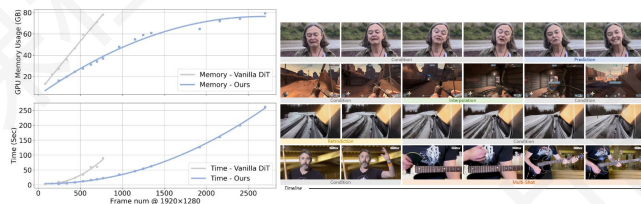
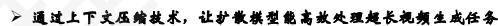
哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY



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5) **前向预测**

时刻1: 已有视频 时刻2: 已有视频 时刻3: 生成视频

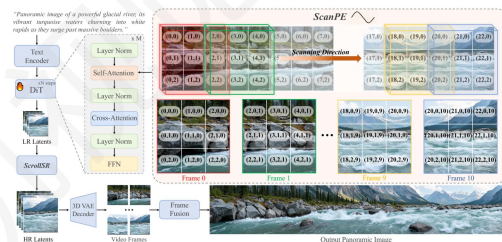
反向预测

时刻1: 生成视频 时刻2: 已有视频 时刻3: 已有视频

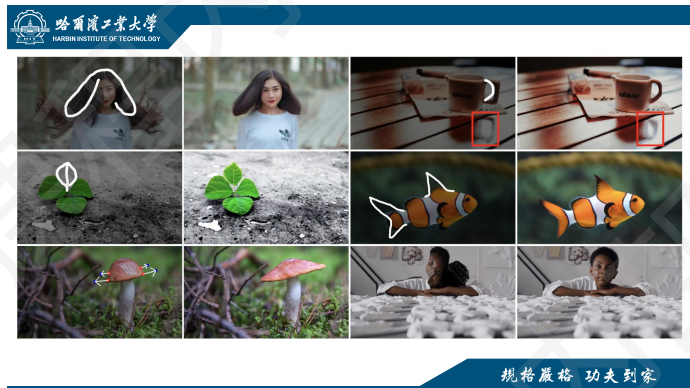
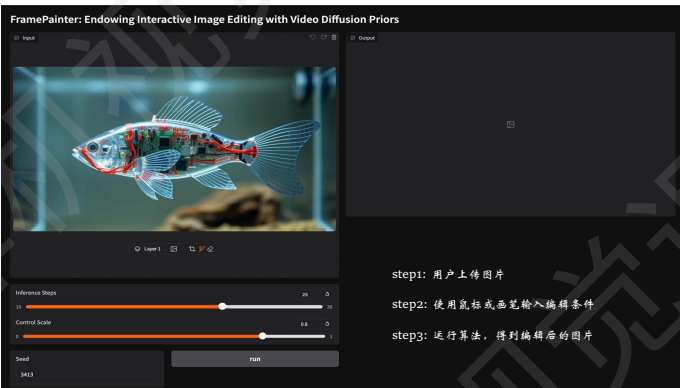
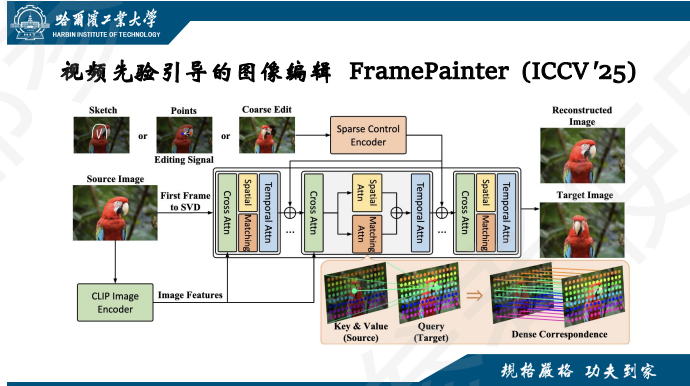
插入生成视频

时刻1: 已有视频 时刻2: 生成视频 时刻3: 已有视频

视频先验引导的长图像生成 ScrollScape (ArXiv'26)



規格嚴格 功夫到家



The End
THANKS FOR YOUR ATTENTION

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